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Professional Profile

I am an M.S. candidate in the AI-Embedded Software on Chip Lab at Kyungpook National University, advised by Prof. Daejin Park. My research focuses on **real-time embedded systems and timing-predictable automotive software for Software-Defined Vehicles (SDVs)**. I work across the hardware–software boundary: register-level commit timing on FPGAs, on-chip reliability diagnostics on automotive MCUs, mixed-criticality command scheduling, and reproducible distributed ECU-in-the-Loop (EIL) co-simulation. I validate this work on real hardware such as the Infineon AURIX TC375 and the Xilinx Zynq-7020. As first author, I have a paper accepted at **EMSOFT 2026** (to appear in **IEEE TCAD**), and I have presented work at IEEE IECON, IPFA, ICAIIC, ICCE-TW, ISPACS, and VNC.

Research Interests: Real-Time Embedded Systems · Timing-Predictable Automotive Software · Software-Defined Vehicles · Mixed-Criticality Control · ECU-in-the-Loop Co-Simulation · Hardware/Software Co-Design · Embedded AI

Education

Kyungpook National University (KNU), Daegu, Korea — Mar 2025 – Present (expected Feb 2027)
M.S. in Electronic and Electrical Engineering, School of Electronic and Electrical Engineering
Advisor: Prof. Daejin Park | AI-Embedded Software on Chip Lab
Thesis (in progress): *On the Hardware–Software Gap in Runtime Control* (working title)

Kyungpook National University (KNU), Daegu, Korea — Mar 2019 – Feb 2025
B.S. in Electronics Engineering, School of Electronics Engineering, College of IT Engineering

Publications

IEEE TCAD (EMSOFT) ×1 **KCI Journal ×2** **Int'l Conference ×6** **Under Review ×1** First author on all

Journal Articles

[J3] **H. Lee** and D. Park, "JSEC: Jitter-to-Step Execution Contracts for Reproducible Distributed ECU-in-the-Loop Co-Simulation," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, to appear.
EMSOFT 2026 Journal Track (ACM/IEEE Int. Conf. on Embedded Software); presented at ESWEEK 2026, Barcelona, Spain, Oct. 2026.

[J2] **H. Lee** and D. Park, "Real-Time Depth Map Using Stereo Vision and Convolutional Filtering," *Journal of the Korea Institute of Information and Communication Engineering (JKIICE)*, vol. 29, no. 9, Sep. 2025. (KCI)

[J1] **H. Lee** and D. Park, "Auto Parking Assistance Control using Human-Activity Feedback-based Embedded Software Emulation," *Journal of the Korea Institute of Information and Communication Engineering (JKIICE)*, vol. 28, 2024. (KCI)

Manuscripts Under Review

[U1] **H. Lee** and D. Park, "Command-Authority Scheduling for Bounded Low-Criticality Service Influence in Mixed-Criticality SDV Control," *IEEE Access*, under review (revised manuscript submitted), 2026.

International Conference Papers

[C6] **H. Lee** and D. Park, "Early Detection of Software-Induced Internal Margin Degradation in Automotive MCUs Using On-Chip ADC Variability," in *Proc. IEEE Int. Symp. on the Physical and Failure Analysis of Integrated Circuits (IPFA)*, Singapore, Jul. 2026.

[C5] **H. Lee**, H. Kim, and D. Park, "Architecture-Aware Neural Compression for TriCore Firmware using Knowledge Distillation," in *Proc. Int. Conf. on Artificial Intelligence in Information and Communication (ICAIIC)*, Tokyo, Japan, Feb. 2026.

[C4] **H. Lee** and D. Park, "Lightweight Empirical Reinforcement Learning Driven Adaptive Super-Twisting Control with Fused Linear-Nonlinear Sliding Surfaces for Embedded Vehicle Control," in *Proc. 51st Annual Conf. of the IEEE Industrial Electronics Society (IECON)*, Madrid, Spain, Oct. 2025.

[C3] **H. Lee** and D. Park, "Advanced Real-Time Performance QEMU Emulated Automotive Control System with a Multithreaded PID Controller," in *Proc. IEEE Int. Conf. on Consumer Electronics – Taiwan (ICCE-TW)*, Kaohsiung, Taiwan, Jul. 2025.

[C2] **H. Lee** and D. Park, "Convolution-Based Depth Map With Shadow Removal Using Cameras for 3D Mapping in Autonomous Vehicle Driving," in *Proc. IEEE Int. Symp. on Intelligent Signal Processing and Communication Systems (ISPACS)*, Kaohsiung, Taiwan, Dec. 2024.

[C1] **H. Lee** and D. Park, "Multi-Camera Interoperable Emulation Framework Using Embedded Edge-Cloud AI Computing for Autonomous Vehicle Driving," in *Proc. IEEE Vehicular Networking Conf. (VNC)*, Kobe, Japan, May 2024. (Poster)

Thesis

[T1] **H. Lee**, "On the Hardware–Software Gap in Runtime Control" (working title), M.S. thesis, Kyungpook National University, in progress (expected Feb. 2027).

Research Experience

Binding-Time Analysis of the Hardware–Software Gap in Runtime Control 2026 – Present

M.S. Thesis

- Framing hardware–software partitioning as a choice of *binding time*, the instant a control decision becomes irrevocable, and studying its mismatch with the rate at which the controlled environment changes.
- Root-causing fragmented control performance on a Xilinx Zynq-7020 at the level of clock cycles and registers, using paired HW/SW implementations, a bit-exact datapath model, and predictions fixed before every measurement.
- Showed that committing a result 160 ns later separated a stable loop from an unstable one, and that the *order* of asynchronous software delays, not only their distribution, decided stability.
- Designed controllers that account for their own commit delay, and a judgment-to-use atomic commit that closes gaps software monitors leave open. Extending the analysis to asynchronous AI workloads on homogeneous and heterogeneous embedded hardware.

Command-Authority Scheduling for Mixed-Criticality SDV Control May 2026 – Present

[U1] IEEE Access (under review)

- Proposed Command-Authority Scheduling (CAS), which converts low-criticality service outputs into role-constrained authority claims and allocates authority tokens under channel-wise ceilings issued by the command domain at every tick.
- Synthesized the ceilings at every tick from an identified plant model and a response envelope built from ISO 15622 acceleration limits, the RSS safe distance, and UN R79 lateral limits. A direction-dependent freshness rule keeps stale evidence of a hazard but not of a clear path.
- On an AURIX TC375 ECU-in-the-loop platform with real-time CARLA, CAS was the only service-integrating policy that never left the envelope. Against a service-free baseline it cut RSS-distance violations by 97% and safety-governor interventions by 90%, with a worst-case receive-to-transmit time of 167 μs (1.7% of the 10 ms period).

Reproducible Distributed ECU-in-the-Loop Co-Simulation (JSEC) Oct 2025 – Oct 2026

[J3] EMSOFT 2026 / IEEE TCAD

- Built a distributed CARLA–Simulink (14-DOF vehicle dynamics)–AURIX TC375 co-simulation platform with bi-directional UDP closed-loop communication. It first served as a ride-comfort testbed for hybrid imitation-learning and PPO control evaluated under ISO 2631.
- Proposed JSEC (Jitter-to-Step Execution Contract), which treats network jitter as an explicit input. Sequence-indexed admission and jitter-to-step mapping turn ECU event arrivals into fixed-step plant advancement, with bounded catch-up execution and deterministic road-input interpolation.
- Achieved a mean R^2 of 0.947 across repeated runs (vs. 0.410 for conventional asynchronous co-simulation) on heave, roll, pitch, and suspension-force responses, with only 0.79–1.14% Simulink execution-time overhead.

On-Chip ADC Variability for Early Detection of Software-Induced Margin Degradation Dec 2025 – Jul 2026

[C6] IPFA 2026

- Showed on an Infineon AURIX TC375 that software-induced internal stress first appears as increased *variability* of on-chip ADC outputs, while ISR runtime and ADC means stay within normal ranges.
- Built a lightweight in-field diagnostic that uses ISR execution time as a load indicator and a formal 3σ threshold to separate latent stress from environmental noise such as external motor loads, with no extra sensors or hardware changes.

Architecture-Aware Neural Firmware Compression via Knowledge Distillation Jun 2025 – Feb 2026

[C5] ICAIIC 2026

- Designed a TriCore-aware lossless compression framework for SDV ECU firmware. A Transformer teacher

distills TriCore binary patterns into a lightweight GRU student that predicts the next byte and drives an arithmetic coder.
- Reached 1.524 bpb (19.05% of the original size) on TC375 firmware, about 38% better than general-purpose compressors, with a 0.25 MB decoding footprint (about 94% lower). Python-vs-C cross-validation confirmed bit-exact, deterministic decoding.

Lightweight Adaptive Sliding-Mode Control with Empirical Reinforcement Learning Jan 2025 – Jun 2025
[C4] IECON 2025
- Developed a lightweight empirical RL method that tunes Super-Twisting Algorithm (STA) gains online through rule-based gradient approximation, avoiding heavy neural-network training on embedded targets.
- Designed a hybrid sliding surface that blends linear and nonlinear components with sigmoid weighting, together with a learning-friendly reward, to balance tracking accuracy against control-input smoothness across disturbance scenarios.

Multithreaded PID-Based Real-Time Automotive Control Architecture Sep 2024 – Jan 2025
[C3] ICCE-TW 2025
- Split sensing, computation, and actuation across parallel threads and precomputed multiple control candidates, so external disturbances (inclines, loads, road surfaces) are compensated with minimal transient deviation.
- Implemented the controller in portable C and validated it in a QEMU–Simulink co-simulation with a PWM-to-voltage DC-motor actuator model.

Camera-Only Depth Mapping for 3D Perception in Autonomous Driving Apr 2024 – Sep 2025
[C2] ISPACS 2024, [J2] JKIICE 2025
- Generated LiDAR-free depth maps from stereo cameras using calibration and rectification, a custom convolution kernel for pixel-level matching uniqueness, and shadow and background removal through mean-shift filtering and edge processing.
- Fused left and right disparities into object-centered minimal-information depth maps and 3D point clouds (Open3D), reaching real-time performance suitable for edge devices without deep-learning models.

Multi-Camera Edge-Cloud ADAS Emulation and Driver-Adaptive Parking Assistance Jan 2024 – Nov 2024
[C1] IEEE VNC 2024, [J1] JKIICE 2024
- Built an interoperable emulation framework that implements Smart Cruise Control and Lane Keeping Assist with YOLO-based detection, camera calibration, real-time TCP/IP video streaming, remote vehicle control, and over-the-air algorithm updates.
- Developed a four-camera auto-parking assistant that adapts control sensitivity in real time to the driver's behavior and the parking environment, settling in about 1 s.

Technical Skills

Programming	C, C++, Python, MATLAB
Embedded HW	Infineon AURIX TC375 (TriCore), Xilinx Zynq-7020 (FPGA), STM32
Real-Time & OS	ISR/timing analysis, multithreaded control, QEMU virtualization, Linux
Simulation	MATLAB/Simulink (14-DOF vehicle dynamics), CARLA, distributed UDP co-simulation
AI/ML	PyTorch, reinforcement learning (PPO), imitation learning, knowledge distillation (Transformer/GRU)
Vision	OpenCV, stereo vision & calibration, YOLO, Open3D

Languages

Korean – Native
English – Business Competence

References

Prof. Daejin Park (M.S. Advisor)
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